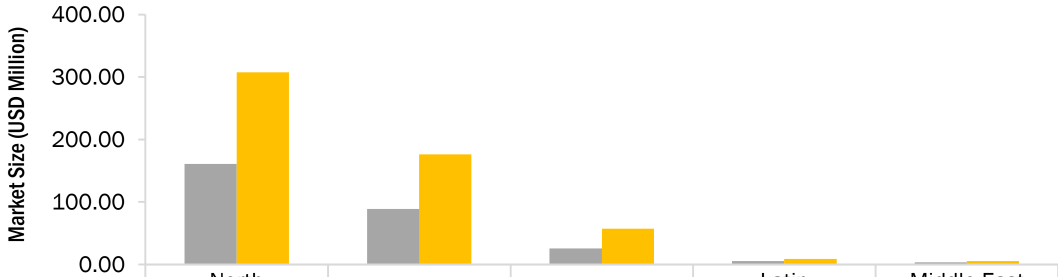
10.1 INTRODUCTION

The global microbiome sequencing services market has been segmented into five major regional segments, namely, North America, Europe, the Asia Pacific, Latin America, and the Middle East & Africa. In 2022, North America dominated the market with the largest share of 56.9%, followed by Europe (31.1%). Factors such as the established research infrastructure, technological advancements, demand for precision medicine, supportive regulatory environment, robust biotechnology and healthcare industry, and availability of funding and investment are driving the growth of the microbiome sequencing services market in North America.

The microbiome sequencing services market in the Asia Pacific is estimated to register the highest growth rate of 17.3% during the forecast period, primarily due to the increasing awareness of microbiome health, the emerging healthcare and biotechnology industry, government support, and funding potential for personalized medicine.

FIGURE 25 MICROBIOME SEQUENCING SERVICES MARKET, BY REGION, 2023 VS. 2028 (USD MILLION)



	North America	Europe	Asia Pacific	Latin America	Middle East and Africa
2023	161.06	88.70	25.73	5.26	3.55
2028	307.60	176.01	57.19	8.88	5.55
CAGR (2023–2028)	13.8%	14.7%	17.3%	11.1%	9.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 87 MICROBIOME SEQUENCING SERVICES MARKET, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	125.75	142.35	161.06	182.39	206.72	235.12	268.35	307.60	13.8%
Europe	68.21	77.80	88.70	101.22	115.60	132.49	152.38	176.01	14.7%
Asia Pacific	18.81	22.01	25.73	30.07	35.16	41.22	48.46	57.19	17.3%
Latin America	4.29	4.75	5.26	5.82	6.44	7.15	7.96	8.88	11.1%
Middle East and Africa	2.97	3.25	3.55	3.88	4.23	4.63	5.06	5.55	9.3%
Total	**220.03**	**250.17**	**284.31**	**323.39**	**368.16**	**420.61**	**482.21**	**555.24**	**14.3%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.2 NORTH AMERICA

North America has emerged as a leading market for microbiome sequencing services, driven by a combination of factors. One of the key drivers is the well-established healthcare and biotechnology industry in the region, with advanced infrastructure, cutting-edge research institutions, and a strong ecosystem of biotechnology companies and CROs specializing in microbiome research. The presence of top academic institutions, research hospitals, and private companies in North America with dedicated microbiome research programs and expertise has contributed to the growth of the microbiome sequencing services market. In 2022, the National Library of Medicine released a research paper that discussed the correlation of gut microbiome diversity and abundance with Gray Matter Volume (GMV) in older adults with depression.

Another factor driving North America's leadership in this market is the high level of awareness among healthcare professionals and the general population about the importance of the human microbiome in health and disease. There has been a growing recognition of the role of the microbiome in various conditions, including gastrointestinal disorders, metabolic diseases, neurodegenerative diseases, and immune-related disorders. This increased awareness has led to a rising demand for microbiome sequencing services to better understand the complex interactions between the human microbiome and host health, as well as to develop targeted interventions and therapies.

Despite the positive drivers, there are also some restraints that can impact the microbiome sequencing services market in North America. One of the major challenges is the high cost associated with certain microbiome sequencing services, which can limit the adoption of these services for some customers, especially smaller research institutions or clinics with limited budgets. Regulatory and ethical considerations related to the use of human microbiome data, such as data privacy and consent issues, can also pose challenges in the market. However, there also are significant opportunities for growth in the North American microbiome sequencing services market. Advancements in sequencing technologies, data analysis, and bioinformatics are enabling more accurate and cost-effective microbiome sequencing, which can propel the adoption of these services. Increasing research collaborations between academia, industry, and government institutions in North America can also lead to novel discoveries and applications of microbiome sequencing, thus creating new market opportunities. The rising demand for personalized medicine approaches, where understanding an individual's microbiome can inform tailored treatment plans, presents a significant growth opportunity for microbiome sequencing services.